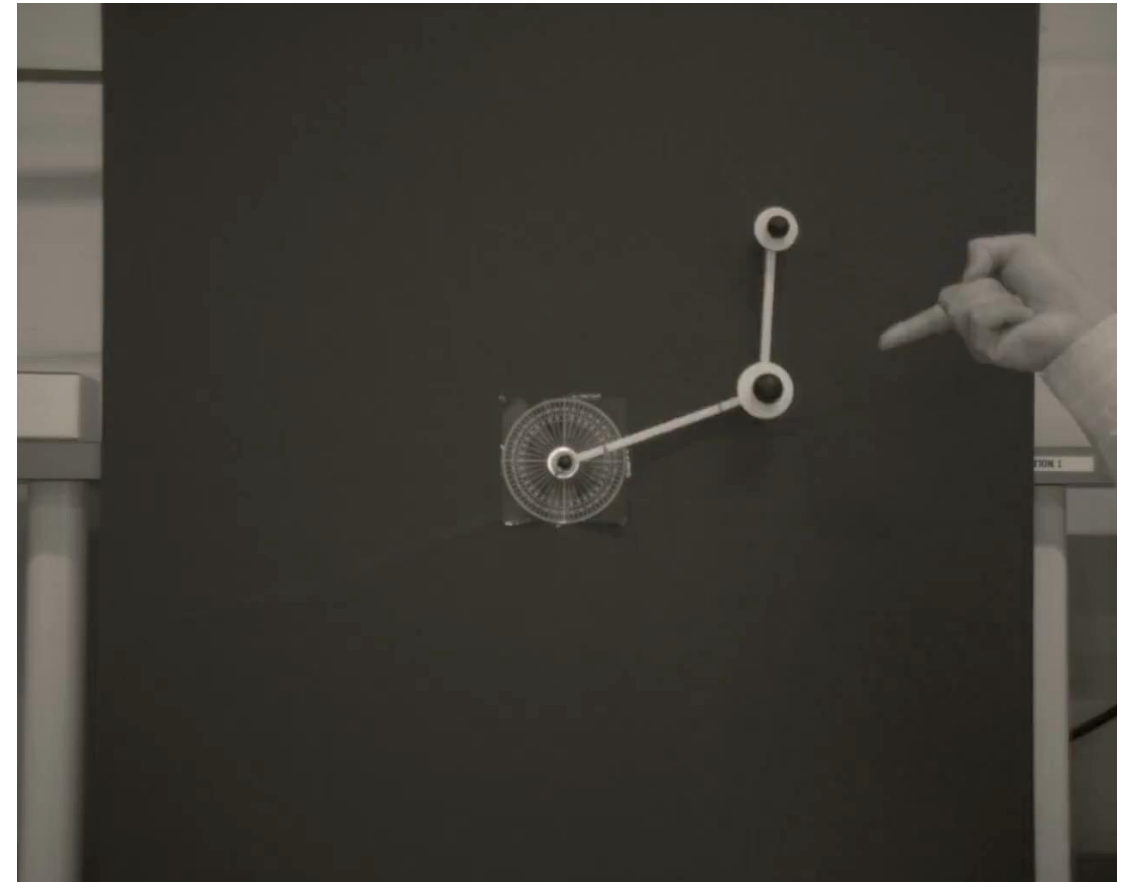


Double pendulum: motion, energy and measurement

Lecture 04 · MIE 402 · Fall 2026

A physical explanation of the model and the Lab 3 workflow.

Core lesson plus derivations for discussion and review.



The work students will do

Measure the two links and the masses attached to them.

Record at least four releases and track both moving bob centers.

Convert saved positions into angles and compare the measured motion with a simulation.

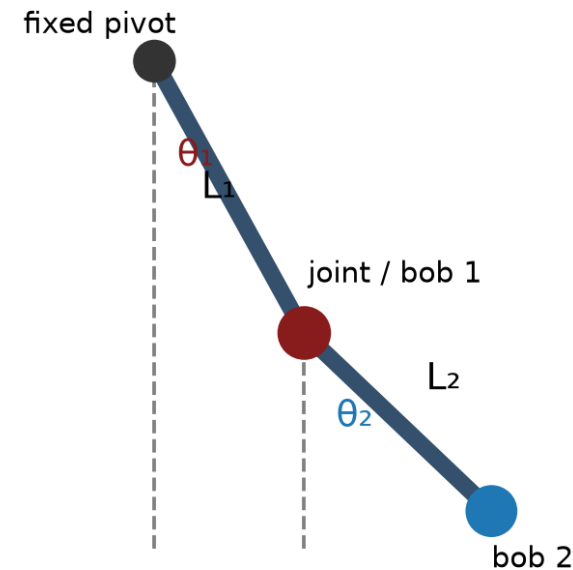
Keep the MP4, ExpData MAT file, settings and trial log for every release.

Two links and two absolute angles

θ_1 [rad]: orientation of the upper link. θ_2 [rad]: orientation of the lower link. Both start from the downward vertical.

L_1 [m]: fixed pivot to joint. L_2 [m]: joint to lower-bob center. Positive angles turn each downward link toward the right.

The joint angle is $\theta_2 - \theta_1$. The lower-link absolute angle θ_2 is measured from the vertical, even while the upper link moves.



Angles are absolute from downward vertical; θ_2 is not $\theta_2 - \theta_1$.

Masses and lengths in the model

m_1, m_2 [kg]: masses of the upper and lower uniform bars.

M_1 [kg]: lumped mass at the moving joint. M_2 [kg]: lumped mass at the lower bob.

L_1 [m]: fixed pivot to moving joint. L_2 [m]: moving joint to lower-bob center.

Assign each washer and attachment to one component only. Use measured values, not the pre-lab example.

Angular velocity: what ω_1 and ω_2 mean

$$\omega_1 = \dot{\theta}_1 = \frac{d\theta_1}{dt} \quad \omega_2 = \dot{\theta}_2 = \frac{d\theta_2}{dt}$$

$\omega_1 = d\theta_1/dt$ [rad/s]: angular velocity of the upper link relative to the fixed vertical.

$\omega_2 = d\theta_2/dt$ [rad/s]: angular velocity of the lower link relative to the fixed vertical. The joint's relative angular velocity is $\omega_2 - \omega_1$.

α_1 and α_2 [rad/s²]: rates of change of ω_1 and ω_2 . Here t is time [s], $i = 1$ or 2 , and a dot denotes a time derivative.

The sign gives rotation direction. Angular speed is $|\omega_i|$. These instantaneous velocities are not oscillation frequencies in hertz.

Position symbols and coordinate convention

(x_1, y_1) [m]: position of the joint / first bob relative to the fixed pivot.

(x_2, y_2) [m]: position of the lower bob relative to the same fixed pivot.

The origin is the fixed pivot. x increases rightward and y increases upward. A point below the pivot has negative y .

Use meters in the energy model. Later, convert the analyzer's saved millimeter coordinates to meters when needed.

Positions follow from the geometry

$$x_1 = L_1 \sin \theta_1 \quad y_1 = -L_1 \cos \theta_1$$

$$\begin{aligned} x_2 &= L_1 \sin \theta_1 + L_2 \sin \theta_2 \\ y_2 &= -L_1 \cos \theta_1 - L_2 \cos \theta_2 \end{aligned}$$

Choose the fixed pivot as (0,0), with x positive right and y positive upward.

Bob 2 moves because both links move. Its coordinates contain a contribution from each link.

Why we use energy to derive the model

T [J] is kinetic energy, the energy of motion. V [J] is gravitational potential energy, determined by mass height.

$E = T + V$ [J] is total mechanical energy. $\mathcal{L} = T - V$ [J] is the Lagrangian, the quantity used to derive the motion equations.

The energy method includes joint constraints without introducing every unknown internal force.

Baseline assumptions: rigid planar links, uniform bars, fixed pivot and no friction or air drag.

Height symbols before calculating potential energy

h_1 and h_2 [m]: upward displacement of bob 1 and bob 2 from their positions when both links hang vertically downward.

h_{G1} and h_{G2} [m]: upward displacement of the centers of the upper and lower bars from that same reference configuration.

The subscript G denotes a bar's center of mass. For a uniform bar, this center is halfway along its length.

For a generic mass m [kg] raised by h [m], the energy increase is mgh [J]. $g \approx 9.81 \text{ m/s}^2$ is gravitational acceleration.

Potential energy begins with height

$$h_{G1} = \frac{L_1}{2} (1 - \cos \theta_1)$$

$$h_1 = L_1 (1 - \cos \theta_1)$$

$$h_{G2} = h_1 + \frac{L_2}{2} (1 - \cos \theta_2)$$

$$h_2 = h_1 + L_2 (1 - \cos \theta_2)$$

Set $V = 0$ when both links hang vertically downward.

Raising any mass by h increases its gravitational potential energy by mgh .

The center of a uniform bar lies halfway along it. The entire lower assembly also rises when link 1 rotates.

The gravity coefficients G_1 and G_2

G_1 [N·m = J]: coefficient collecting the gravitational effect of all masses raised when θ_1 changes.

G_2 [N·m = J]: coefficient collecting the gravitational effect of the lower bar and bob when θ_2 changes.

These capital G symbols are energy coefficients. The G in a center-of-mass subscript, such as h_{G2} , only labels a location.

We will group the four contributions to V using these two coefficients. They depend on measured masses, lengths and g.

Potential energy of all four components

$$V = G_1(1 - \cos \theta_1) + G_2(1 - \cos \theta_2)$$

$$G_1 = \left(\frac{m_1}{2} + M_1 + m_2 + M_2\right)gL_1$$

$$G_2 = \left(\frac{m_2}{2} + M_2\right)gL_2$$

Add the energy of both bars and both lumped masses.

G_1 and G_2 are constants with units N·m, or joules. They collect mass, gravity and distance.

A larger mass or longer lever arm increases the energy needed to raise the system.

Gravity produces a restoring torque

$$Q_{g,i} = -\frac{\partial V}{\partial \theta_i} = -G_i \sin \theta_i$$

$Q_{g,i}$ [N·m]: generalized gravitational torque associated with angle θ_i , for $i = 1$ or 2 . Subscript g denotes gravity.

$\partial V / \partial \theta_i$ means the change in potential energy per change of θ_i while holding the other angle fixed.

For a small positive angle, the negative sign means gravity acts to reduce that angle. Each G_i includes the masses raised by that angle.

Speed and angle symbols in the bob energy

v_1 and v_2 [m/s]: magnitudes of the translational velocities of bob 1 and bob 2, measured relative to the fixed laboratory.

$T_{\text{bob } 1}$ [J]: kinetic energy of bob 1. In the generic point-mass formula $T = \frac{1}{2}Mv^2$, M is that point's mass [kg] and v is its speed [m/s].

$\Delta = \theta_1 - \theta_2$ [rad]: difference between the two absolute link angles. It determines the angle between the two velocity contributions.

ω_1 and ω_2 are signed angular velocities [rad/s]. Multiplying a link length by angular velocity gives a tangential velocity contribution.

Kinetic energy of a moving bob

$$T_{bob\ 1} = \frac{1}{2} M_1 L_1^2 \omega_1^2$$

$$v_2^2 = L_1^2 \omega_1^2 + L_2^2 \omega_2^2 + 2L_1 L_2 \omega_1 \omega_2 \cos \Delta$$

For a point mass, $T = \frac{1}{2} Mv^2$. Bob 1 moves on a circle about the fixed pivot.

Bob 2 has the velocity of the moving joint plus its motion relative to that joint.

Squaring that sum produces a cross term. This is the origin of kinetic coupling.

Symbols for a moving bar

For a generic uniform bar, m [kg] is its mass and L [m] is its length.

I_{center} and I_{end} [$\text{kg}\cdot\text{m}^2$]: mass moments of inertia about axes through the bar center and end, perpendicular to the motion plane.

v_{G2} [m/s]: translational speed of the lower bar's center of mass. I_{G2} [$\text{kg}\cdot\text{m}^2$] is its inertia about that center.

$T_{\text{bar } 2}$ [J]: lower-bar kinetic energy. It includes translation of the center and rotation about the center.

A bar also carries kinetic energy

$$I_{center} = \frac{mL^2}{12} \quad I_{end} = \frac{mL^2}{3}$$
$$T_{bar\ 2} = \frac{1}{2}m_2v_{G2}^2 + \frac{1}{2}I_{G2}\omega_2^2$$

A uniform bar rotating about its end has moment of inertia $mL^2/3$.

The upper bar rotates about the fixed pivot. The lower bar also translates because its upper joint moves.

Include both translation of the center and rotation about the center for the lower bar.

The inertia coefficients I_{11} , I_{22} and C

I_{11} [$\text{kg}\cdot\text{m}^2$]: coefficient multiplying ω_1^2 . It includes the lower assembly because the upper link carries it through space.

I_{22} [$\text{kg}\cdot\text{m}^2$]: coefficient multiplying ω_2^2 . It is the lower assembly's inertia about the moving joint.

C [$\text{kg}\cdot\text{m}^2$]: coefficient of the product $\omega_1\omega_2 \cos \Delta$. It measures how both rotations contribute to the same moving masses.

T_{cross} [J] names that product term in total kinetic energy. Its sign can change, while the total kinetic energy remains nonnegative.

Total kinetic energy

$$T = \frac{1}{2}I_{11}\omega_1^2 + \frac{1}{2}I_{22}\omega_2^2 + C \cos \Delta \omega_1\omega_2$$

$$I_{11} = \left(\frac{m_1}{3} + M_1 + m_2 + M_2\right)L_1^2$$

$$I_{22} = \left(\frac{m_2}{3} + M_2\right)L_2^2$$

$$C = \left(\frac{m_2}{2} + M_2\right)L_1L_2$$

The first two terms account for motion associated with each angular speed.

The cross term accounts for the fact that both rotations contribute to the lower assembly's velocity.

I_{11} , I_{22} and C all have units $\text{kg}\cdot\text{m}^2$. They depend only on the apparatus masses and lengths.

Meaning of the coupling term

$$T_{cross} = C \cos(\theta_1 - \theta_2) \omega_1 \omega_2$$

$\Delta = \theta_1 - \theta_2$ measures the difference between the link orientations.

For aligned links moving in the same sense, the lower-bob velocity contributions reinforce each other.

Opposite angular speeds can partly cancel those contributions. The cross term may be negative, while total T remains nonnegative.

At $\Delta = 90^\circ$, this cross term is zero at that instant. Other dynamic coupling terms can still remain.

Torques and the rate of energy loss

Q_1 and Q_2 [N·m]: generalized nonconservative torques associated with θ_1 and θ_2 , such as the effects of pivot friction or air drag.

Gravity is already included in V . These Q_1 and Q_2 differ from the gravitational torques $Q_{g,1}$ and $Q_{g,2}$.

$Q_i\omega_i$ [W] is the mechanical power associated with coordinate i . The total dE/dt [J/s = W] is the rate of change of mechanical energy.

At the moving joint, friction acts on relative rotation. Its effect must be included consistently in both generalized torques.

Energy during real and ideal motion

$$E = T + V \quad \frac{dE}{dt} = 0 \text{ (ideal model)}$$

$$\frac{dE}{dt} = Q_1\omega_1 + Q_2\omega_2 \text{ (with nonconservative torques)}$$

At release from rest, T is zero and all initial mechanical energy is potential energy.

As masses descend, some potential energy becomes kinetic energy. Both links exchange energy through their connection.

The ideal simulation conserves total E. Real pivot friction and air drag remove mechanical energy.

Energy loss in a recording does not by itself identify which loss mechanism dominates.

How energy gives the equations of motion

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \omega_i} \right) - \frac{\partial \mathcal{L}}{\partial \theta_i} = Q_i \quad i = 1, 2$$

Form $\mathcal{L} = T - V$, then apply the equation once for θ_1 and once for θ_2 .

The first term accounts for the rate of change of generalized momentum. The second accounts for how the energy changes with configuration.

For the ideal baseline, $Q_1 = Q_2 = 0$. A supplied numerical solver advances these equations in time.

The two coupled acceleration equations

$$I_{11}\alpha_1 + C \cos \Delta \alpha_2 = -C \sin \Delta \omega_2^2 - G_1 \sin \theta_1$$

$$C \cos \Delta \alpha_1 + I_{22}\alpha_2 = C \sin \Delta \omega_1^2 - G_2 \sin \theta_2$$

The same I_{11} , I_{22} , C , G_1 and G_2 from the energy expressions appear here.

The terms containing $\sin \Delta$ and squared angular speed arise because the geometry changes during motion.

Both angular accelerations must be solved together at each time step.

What the simulation code actually does

$$q = (\theta_1, \theta_2)^T \quad \alpha_i = \frac{d\omega_i}{dt}$$

$q = (\theta_1, \theta_2)^T$ [rad] is the two-angle column vector. Superscript T means transpose. Inputs also include both angular velocities, masses and lengths.

At the current angles and velocities, calculate the two right-hand sides and the 2×2 inertia matrix.

Solve for α_1 and α_2 , then advance the angles and angular velocities in time.

Students run the supplied solver. A small change in time step should not change the physical conclusion.

Matrix symbols for small oscillations

M_0 [kg·m²]: the constant 2×2 inertia matrix evaluated at the downward equilibrium. Its entries are I_{11} , I_{22} and C .

K [N·m/rad]: the angular restoring-stiffness matrix. Its diagonal entries are G_1 and G_2 , with radians treated as dimensionless in SI.

q is the column of two angles. $\ddot{q}$ is the column of their angular accelerations [rad/s²]. The subscript 0 denotes the equilibrium value.

$\text{diag}(G_1, G_2)$ means a matrix with G_1 and G_2 on its diagonal and zeros elsewhere. The matrix M_0 is distinct from the bob masses M_1 and M_2 .

Small motion and normal modes

$$M_0 \ddot{q} + Kq = 0$$
$$M_0 = [I_{11}, C; C, I_{22}] \quad K = \text{diag}(G_1, G_2)$$

For angles close to zero, $\sin \theta \approx \theta$ and $\cos \Delta \approx 1$. Products of small quantities drop out.

The linear model has two normal modes: one largely moves the links together and the other largely moves them oppositely.

A general small release can contain both modes. The angle trace can therefore contain two frequencies.

How the two mode frequencies are found

$$q(t) = a \cos(\Omega t + \phi)$$

$$\det(K - \Omega^2 M_0) = 0 \quad f = \frac{\Omega}{2\pi}$$

$a = (a_1, a_2)^T$ [rad] contains the two signed modal angle amplitudes. ϕ [rad] is starting phase, and t [s] is time.

Ω [rad/s] is a mode's oscillation angular frequency. f [Hz] is its cycles per second. Ω differs from instantaneous angular velocities $\omega_1(t)$ and $\omega_2(t)$.

$\det$ means determinant. Setting it to zero allows a nonzero amplitude vector a and gives the two values of Ω^2 . Divide each Ω by 2π to obtain f .

Large motion and initial conditions

At large angles, the small-angle approximation loses accuracy and the coupling changes throughout the swing.

Nearby starts can separate over time. Use the same model parameters when comparing them.

Release without a push and record both actual initial angles.

A short complicated trace can show sensitivity, but appearance alone does not prove mathematical chaos.

The four laboratory trials

Trial A: small nonzero angles, roughly aligned links, released from rest.

Trials B–D: at least three larger combinations of initial angles.

Include a pair of similar starts so that you can compare sensitivity.

For every trial log both initial angles, masses, lengths, capture frame rate and filenames.

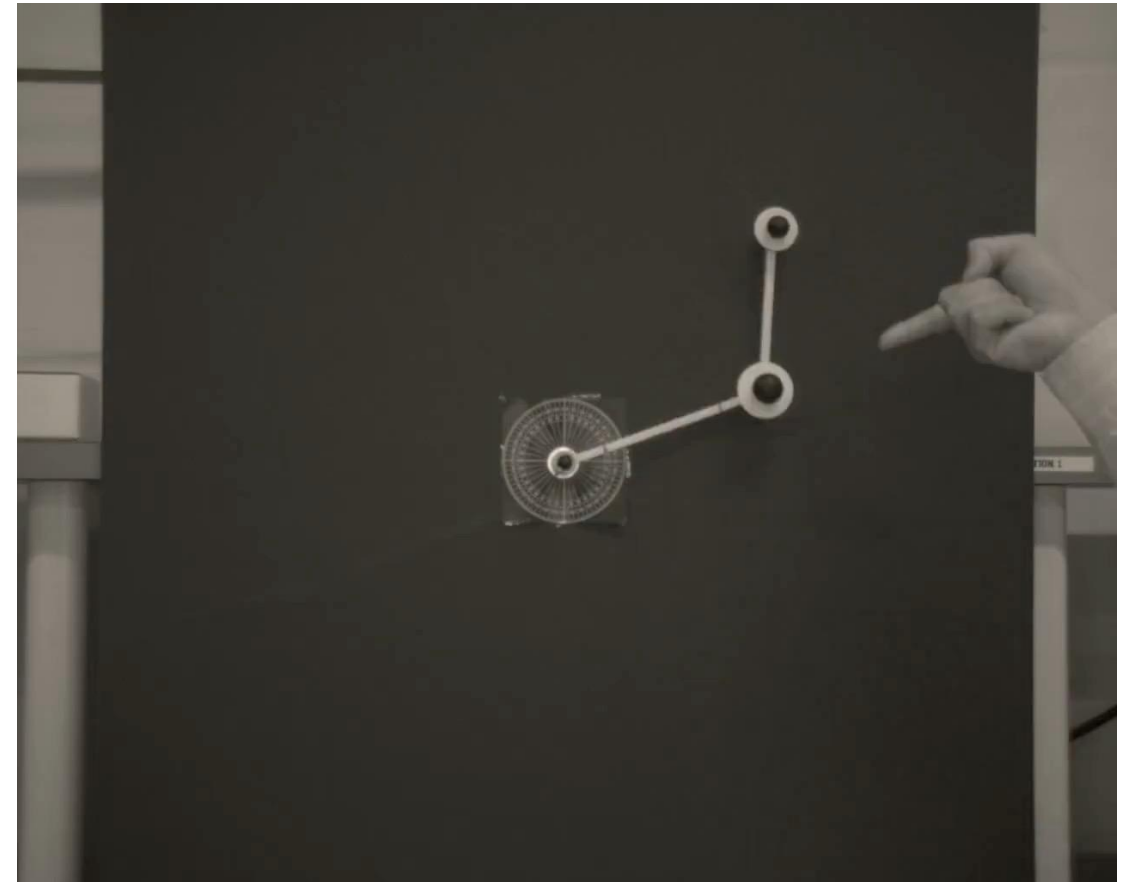
What the camera records

The camera records a sequence of images. The analyzer later extracts positions from them.

Use a dark background, bright bobs, sharp focus and enough framing for the whole motion.

The lab setup uses 1280×1024 and nominal 1000 capture frames/s. Record the actual setting.

Capture frame rate and MP4 playback rate can differ. Enter the capture rate in the analyzer.



Tracking both bobs

Select Double Pendulum in Pendulum Motion Analyzer V4.

Calibrate with a known distance and enter the actual capture frame rate.

Mark the fixed pivot and both bob centers. Inspect early, middle and late frames.

Use Quick Review to find problems, then Analyze All Frames and save results. Missing positions or a marker stuck on a fixed feature require correction.

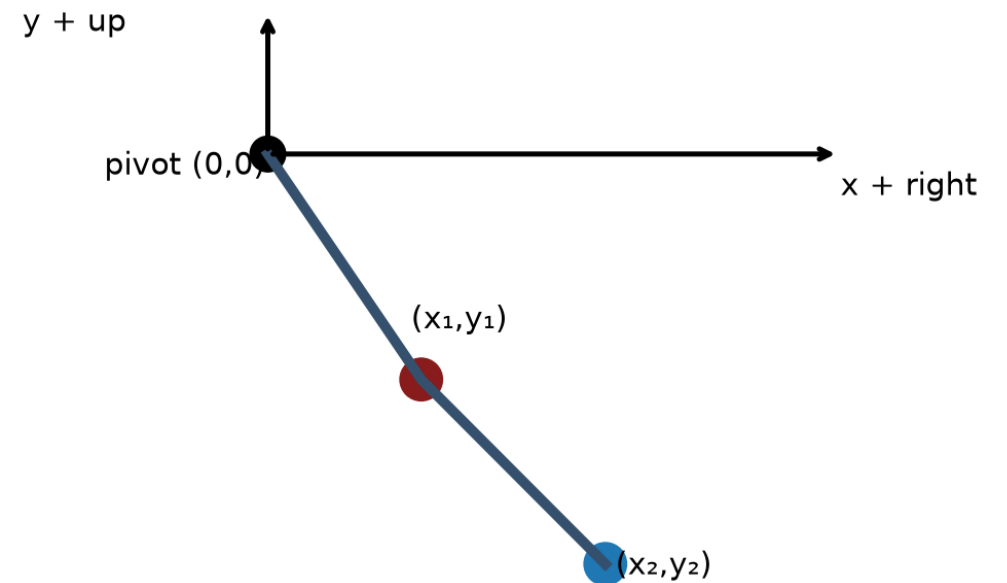
The coordinates in ExpData

Columns: time [s], x_1 [mm], y_1 [mm], x_2 [mm], y_2 [mm].

Coordinates are relative to the fixed pivot, with y positive upward.

The vector from bob 1 to bob 2 is $(x_2 - x_1, y_2 - y_1)$.
This vector gives the lower-link direction.

Inspect the saved MAT file before leaving the lab.
Keep the original MP4.



Positions become the two absolute angles

$$\theta_1 = \text{unwrap}(\text{atan2}(x_1, -y_1))$$

$$\theta_2 = \text{unwrap}(\text{atan2}(x_2 - x_1, -(y_2 - y_1)))$$

atan2 uses both vector components to identify the correct quadrant.

A downward link should give zero angle. Verify the sign using a known video frame.

Unwrapping removes artificial jumps at $\pm\pi$ when tracking continuous motion.

Sampling symbols before the FFT equation

FFT means Fast Fourier Transform. It computes frequency components from an equally spaced sequence of angle measurements.

f_s [Hz]: sample rate of the analyzed angle data. N_s : number of samples used. $T_r = N_s/f_s$ [s]: record duration used for FFT bin spacing.

Δf [Hz]: spacing between neighboring FFT frequency bins. This Δf is different from the angle difference $\Delta = \theta_1 - \theta_2$.

If the analyzer uses every tenth camera frame, f_s is one tenth of the capture frame rate. Use the time column to verify the actual sample spacing.

Angular velocity and frequency analysis

$$\omega_i = \frac{d\theta_i}{dt} \quad \Delta f = \frac{f_s}{N_s} = \frac{1}{T_r}$$

Angular velocity is the slope of the angle trace. Differentiation also amplifies tracking jitter.

FFT identifies frequency components over a finite record. Remove the mean before analysis.

Small motion may show two modal peaks. Large motion can have broader or changing spectral content.

State record length and analysis choices. A single global sine fit may be inappropriate.

Measured motion and the model

Use measured masses, lengths and starting conditions in the simulation.

Overlay both measured angles with their predictions and compare early timing and shape.

For a large-angle trial, explain how sensitive long-time agreement is to starting conditions.

Discuss release push, calibration, friction, air drag and motion out of the camera plane.

Pre-Lab 3 and the laboratory

Run the supplied notebook and answer six questions in one Word or PDF file. No student programming is required.

Compare one small release with two large releases separated by 1° at the start.

Read angle traces, their separation and the spectra. Explain the role of energy and coupling.

Use the pre-lab to plan measurements. Replace illustrative masses and lengths with your measured apparatus values.

The practical sequence in the lab

Measure and label the parts. Decide which mass belongs in each model parameter.

Set the camera and calibration, then record the four releases.

Review pivot and bob markers throughout each recording. Analyze all frames and save ExpData.

Generate two angle traces, two angular-velocity traces and spectra. Review one complete trial with the GTA.

Before leaving the laboratory

Save four trial labels and their measured initial conditions.

Record masses and lengths with units, capture frame rate and calibration distance.

Keep raw MP4 and ExpData MAT files for each trial.

Confirm that the complete analysis opens on your own computer and that the plotted trajectories are physically plausible.

Questions for discussion

If both links start at the same angle, must they remain aligned? Explain using the coupling.

Why does the lower assembly contribute to I_{11} as well as I_{22} ?

How would added lower-bob washers change kinetic and potential energy coefficients?

What observation distinguishes bad tracking from real complex motion?

Reference: remaining symbols and units

h_1, h_2 [m]: rise of each bob above its lowest position. h_{G1}, h_{G2} [m]: rise of each bar center. $g = 9.81 \text{ m/s}^2$ is gravitational acceleration.

Q_1, Q_2 [N·m]: generalized torques from effects excluded from V , such as friction. $Q_i \omega_i$ has units of power [W].

$q = (\theta_1, \theta_2)^T$ is the angle vector. a contains the two modal angle amplitudes; ϕ is the starting phase [rad].

f_s [Hz]: sampling rate of the analyzed data. N_s : number of samples. $T_r = N_s/f_s$ [s]: record duration for FFT bin spacing.

Δf [Hz]: spacing between FFT frequency bins. A longer record reduces Δf ; zero-padding alone does not improve physical resolution.